



EOG Artifacts Suppression From single channel EEG Signals by VME-GMETV model

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ABSTRACT

Nowadays the usage of portable electroencephalography (EEG) recorded systems are raised for recording of neuro activities of the brain, which are useful in clinical diagnosis such as brain-computer interface (BCI) applications. The eye movements or blinks are unavoidable activities, which reflects the electrooculogram (EOG) artifacts in the recorded EEG signals, which leads deceptive interpretation of the underlying brain state. In this paper a novel framework is proposed as Variational Mode Extraction (VME) method is integrated with Generalized Moreau Envelope Total Variation (GMETV) technique to filter the EOG artifacts. Initially, the contaminated EEG is applied to the VME method which extracts the EOG artifact segment. Then, this EOG artifact segment is given to the GMETV technique and it separates the wanted component of the EEG residue from the EOG artifact segment. The EEG residue is added back to the non EOG artifact segment signal which gives the final denoised EEG. Matlab simulations are done on both simulated and real time EEG databases to evaluate the proposed VME-GMETV technique performance in terms of Relative Root Mean Square Error (RRMSE), Correlation Coefficient (CC) and Mean Absolute Error (MAE). From the experimental results, the proposed technique is outperformed the existing methods by averaged low RRMSE (0.1557), MAE (5.0714e-04) and averaged high CC (0.9695).

1. Introduction

The usage of portable systems are made easy the recording of electroencephalography (EEG) in indoor or outdoor environment especially for brain-computer interface (BCI) applications [1]. Prefrontal EEG recordings are mostly preferred for long-term observations of the patient and these are applied for many applications [2–6]. In general, the artifacts either physiological (due to cardiac and brain neural shrinkages etc.) or non-physiological (electrode pop, Electrical Shift and Linear Trend artifacts etc.) are commonly contaminated in the recorded EEG [7,8]. The eye movements or blinks are unavoidable activities, which reflects the electrooculogram (EOG) artifacts in the recorded EEG signals, which leads deceptive interpretation of the underlying brain state [9]. Since, the EOG artifact overlap with the EEG signals frequency spectrum between 0–12 Hz, their removal has proven to be difficult [10].

The employment of lowpass filters to remove the EOG artifacts are not successful, because of loss of wanted portion along with artifacts [11,12]. The adoption of regression and adaptive filters to remove the artifacts are addressed in [13], but they require the reference signal. Wiener filter is employed to remove the artifacts without reference signal, but it suffers initial calibrations difficulties [14]. Several blind

source separation (BSS) techniques are available to remove the artifacts in the multichannel signals such as canonical correlation analysis (CCA) and independent component analysis (ICA) [15–17]. BSS techniques are combined with decomposition algorithms like discrete/ stationary wavelet transform (DWT/SWT), Singular Spectrum Analysis (SSA), local mean decomposition (LMD), ensemble empirical mode decomposition (EEMD) and Variational Mode Decomposition (VMD) [18–24]. In these [18–24], the single channel signal is decomposed into multiresolution data and it is applied to BSS techniques. Even though, the BSS techniques are good in source separation, there is some needful information is present in the artifact component. Forcing this artifact component to zero leads to loss of needful info, which is undesirable in BCI [21,25].

Recently, the combination of overlap segmented adaptive SSA and adaptive noise canceller (ANC) named as OV-ASSA-ANC [26] technique is used to remove the EOG artifacts from single channel EEG. In this, Ov-SSA is used to identify the artifact, it was applied to ANC as reference input. The main drawback of this technique is setting of overlap percentage is difficult task. In [21], the joint use of SSA-ICA-SWT is addressed to remove the EOG artifacts. Here, the single channel EEG is decomposed into several bands using SSA repeatedly,

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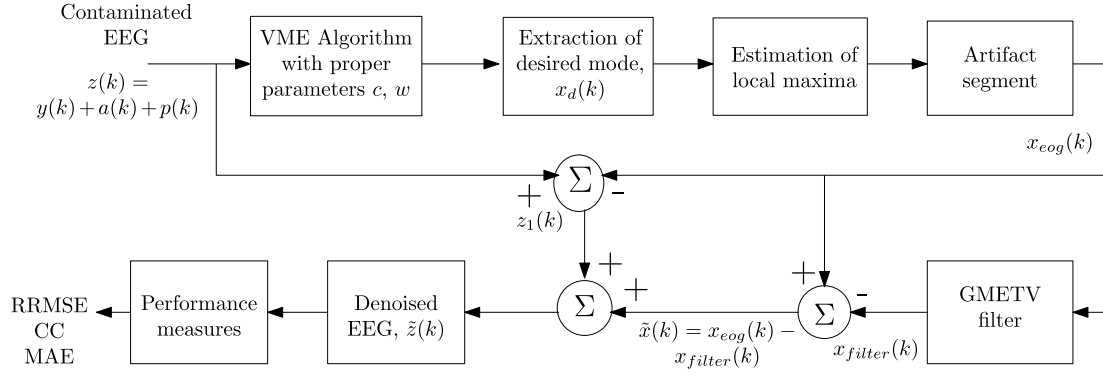


Fig. 1. Proposed VME-GMETV model block diagram.

then ICA is applied on these band of the signals, which will separate the hidden sources as in the form of independent components (IC). The soft thresholding is applied on the artifact IC through the SWT technique. In [27], Variational Mode Extraction (VME)-DWT is proposed to eliminate the EOG artifacts. In this, initially, the EOG artifact segment is identified by VME technique. Then, the identified EOG segment is applied to DWT technique, where the skewness-based thresholding is performed. The existing methods in [21,27], both are used SWT/DWT with thresholding. The performance of these [21,27] methods are depended on the selection of mother wavelet as well as decomposition levels. However, the thresholding introduces loss of wanted component due to non-stationary behavior of EEG signals, which leads the performance degradation [25,28,29].

The authors in [30], conducted a review on the various artifacts removal techniques from multi-channel EEG signals. In [31], Empirical Mode Decomposition (EMD) technique is combined with cross-correlation to suppress EOG artifacts. Here, the single channel contaminated EEG signal is applied to EMD technique, which will decompose into several intrinsic mode functions (IMFs). In order to identify the EOG artifact contaminated IMFs, the statistical parameters, variance, skewness and cross-correlation are calculated for each IMF. However, the EMD performance worsen due to mode mixing problem [32]. In [32], the authors developed a hybrid model, EEMD-kernel Principal Component Analysis (kPCA) to remove the EOG artifacts. The single channel interfered EEG signal is mapped to series of multi-dimensional data named as IMFs using EEMD technique. These IMFs are applied to kPCA, which separates the EOG artifact and the actual EEG signal. The EEMD efficiency is degraded due to edge effect problem [21].

To overcome the above issues, a novel approach is proposed by combining VME technique with Generalized Moreau Envelope Total Variation (GMETV) filter [33], namely VME-GMETV. Initially, the VME technique is employed on signal channel contaminated EEG, which will identify the artifact segment. Later, the GMETV filter is applied on the artifact segment, it will estimate the actual artifact component. This estimated artifact component is subtracted from the VME technique identified artifact segment, which results the wanted component of EEG. The wanted component of EEG is added back to the non artifact segment of EEG signal, which will give the final denoised EEG signals.

The remaining contents of document are organized as follows. The proposed methodology is explained in Section 2. Section 3 represents the existing methods under comparison. The performance metrics and experimental results are presented in Section 4. Section 5 indicates the conclusion of the paper.

2. Methodology

2.1. Proposed VME-GMETV

The implementation of proposed VME-GMETV method is shown in Fig. 1.

The contaminated EEG, $z(k)$ is given to VME algorithm, which estimate the desired mode, $x_d(k)$. For proper estimation of $x_d(k)$, the VME algorithm requires precise selection of compactness co-efficient, c and center frequency, ω . The authors in [34] are suggested higher compactness co-efficient value, but in this paper, we started the c value is 20000 and decreased in the step of 500 up to 1000. Also, we initialized the center frequency, ω is 6 Hz (because the EOG artifact is at lower frequency) and varied in the step of 0.2 Hz. At c is 3500 and ω is 2.8 Hz the estimated desired mode looks good for VME algorithm; hence we fixed these values. After successful extraction of the desired mode, the local maxima is calculated by using threshold formula used in [20,35].

$$T_{th} = \text{median} \frac{|x_d(k)|}{0.6745} \sqrt{(2 \log k)} \quad (1)$$

With the help of T_{th} , the EOG artifact segment, $x_{eog}(k)$ is separated and applied this artifact segment to GMETV filter. Then, the filter estimates the actual artifact, $x_{filter}(k)$, which will be subtracted from the $x_{eog}(k)$. It results the residue of EEG component as follows.

$$\tilde{x}(k) = x_{eog}(k) - x_{filter}(k) \quad (2)$$

The final denoised EEG, $\tilde{z}(k)$ is obtained by the summation of $\tilde{x}(k)$ and $z_1(k)$ as follows.

$$\tilde{z}(k) = \tilde{x}(k) + z_1(k) \quad (3)$$

where $z_1(k) = z(k) - x_{eog}(k)$

2.2. VME

The VME is the modified version of Variational Mode Decomposition (VMD) [24] to extract the specific mode from the given random signal, where the VMD will decompose the all-possible modes. In particular applications like artifact removal, the VMD introduces more computations by decomposing all un-necessary modes. Hence, the VME is better choice than the VMD especially for artifact removal applications.

The VME splits the given input signal into two parts as, desired mode, $x_d(k)$ and undesired mode, $x_{ud}(k)$. The extraction of desired mode is based on the pre-valued center frequency, ω_c bandwidth minimization criteria as proposed in [27,34], which is as follows.

- (i) Hilbert transform is performed on the given input signal as $[(\delta(k) + j/k\pi) * x_d(k)]$, where $\delta(k)$ is dirac delta distribution function.
- (ii) Frequency shifting operation is performed by multiplying the ω_c with the Hilbert transformed output signal as, $[(\delta(k) + j/k\pi) * x_d(k)]e^{-j\omega_c k}$.
- (iii) The bandwidth of the desired mode is defined as the gradient of the shifted input signal's second norm squared.
- (iv) The desired mode bandwidth should as minimum as possible, i.e.,

$$\|\partial_k [(\delta(k) + j/k\pi) * x_d(k)]e^{-j\omega_c k}\|_2^2 \quad (4)$$

(v) There should be proper spectral separation between desired and undesired modes. To maintain this, a filter is considered as follows,

$$F(\omega) = \frac{1}{\rho(\omega - \omega_c)^2} \quad (5)$$

where ρ is regulating parameter.

The spectral overlap between desired and undesired modes are minimized by a penalty equation as follows,

$$J = \|F(k) * x_{ud}(k)\|_2^2 \quad (6)$$

2.3. GMETV

The GMETV [33] is, in contrast to the regular Moreau envelope TV [36], which accommodates a matrix parameter, B . The VME algorithm extracted EOG artifact segment, $x_{eog}(k)$ is evaluated by the GMETV denoising cost function as follows. The general form is

$$C_B(x_{filter}(k)) = \frac{1}{2} \|x_{eog}(k) - x_{filter}(k)\|^2 + \lambda [\|Dx_{filter}(k)\|_1 + inf_v(\|Dv\|_1 + \frac{1}{2} \|B(x_{filter}(k) - v)\|_2^2)] \quad (7)$$

The cost function, $C_B(x_{filter}(k))$ reduces, when matrix parameter, B varies as follows,

$$C_B(x_{filter}(k)) = \begin{cases} \frac{1}{2} \|x_{eog}(k) - x_{filter}(k)\|^2 + \|D_i * x_{filter}(k)\|_1, & \text{for } B = 0 \\ \frac{1}{2} \|x_{eog}(k) - x_{filter}(k)\|^2 + \text{Minimax} - \text{Concave}_b(x_{filter}(k)), & \text{for } B = \sqrt{b}D_i \\ \frac{1}{2} \|x_{eog}(k) - x_{filter}(k)\|^2 + \text{Moreau} - \text{envelope}_b(x_{filter}(k)), & \text{for } B = \sqrt{b}I \end{cases}$$

where b is Moreau-envelope parameter, I is the identity matrix, λ is regularization parameter and D_i is the first-order difference matrix with dimension of $(k-1) * (k)$ [33] and is displayed in the following equation,

$$D_i = \begin{bmatrix} -1 & 1 & & & \\ & -1 & 1 & & \\ & & \ddots & \ddots & \\ & & & -1 & 1 \end{bmatrix} \quad (8)$$

To maximize the $C_B(x_{filter}(k))$ cost function, it should obey

$$BB^T \leq (1/\lambda)I \quad (9)$$

3. Existing algorithms

3.1. Ov-ASSA-ANC

In [26], the authors proposed the overlap segmented adaptive SSA-ANC algorithm to remove the EOG artifacts. In this, the contaminated signal is divided into segments with a small portion of overlapping. The basic steps of adaptive SSA, embedding, singular value decomposition (SVD), grouping and reconstruction operations are performed. Here, the grouping is performed based on the amplitude of contaminated signal, the first one or two reconstructed components are considered. These grouping reconstructed components are considered as artifact signal and fed to ANC as the reference input signal. Then, the ANC estimates the filtered signal. The main drawback of this technique is setting of overlap percentage is difficult task.

3.2. SSA-ICA-SWT

In [21], the joint use of SSA-ICA-SWT is addressed to remove the EOG artifacts. Here, the single channel EEG is decomposed into several bands using SSA repeatedly, then ICA is applied on these band of the signals, which will separate the hidden sources as in the form of

independent components (IC). The soft thresholding is applied on the artifact IC through the SWT technique. Here, the SWT technique calculate the detailed and approximation coefficients of artifact IC and these coefficients are compared with soft thresholding. Those coefficients above soft thresholding are set to zero and remaining coefficients are considered as wanted component.

3.3. VME-DWT

In [27], Variational Mode Extraction (VME)-DWT is proposed to eliminate the EOG artifacts. In this, initially, the EOG artifact segment is identified by VME technique. Then, the identified EOG segment is applied to DWT technique, where the skewness-based thresholding is performed. The existing methods in [21,27], both are used SWT/DWT with thresholding. The performance of these [21,27] methods are depended on the selection of mother wavelet as well as decomposition levels. However, the thresholding introduces loss of wanted component due to non-stationary behavior of EEG signals, which leads the performance degradation [25,28,29].

3.4. EMD - Cross-correlation

In [31], Empirical Mode Decomposition (EMD) technique is combined with cross-correlation to suppress EOG artifacts. Here, the single channel contaminated EEG signal is applied to EMD technique, which will decompose into several intrinsic mode functions (IMFs). In order to identify the EOG artifact contaminated IMFs, the statistical parameters, Variance and Skewness are calculated for each IMF. As per the authors in [31], the IMFs having high values of variance and skewness are considered as EOG artifact contaminated IMF. To conform this another statistical parameter cross-correlation coefficient is calculated between the EOG artifact contaminated IMF and the contaminated EEG signal. If the calculated cross-correlation is greater than the threshold cross-correlation that IMFs are removed and the remaining IMFs are combined, which results the denoised EEG. The threshold cross-correlation value is adjusted such that which gives the low RRMSE value. In this paper the threshold cross-correlation value is considered 0.6, where, the authors in [31] is 0.5.

3.5. EEMD-kPCA

In [32], the authors developed a hybrid model, EEMD-kernel Principal Component Analysis (kPCA) to remove the EOG artifacts. The single channel interfered EEG signal is mapped to series of multi-dimensional data named as IMFs using EEMD technique. These IMFs are applied to kPCA, which separates the EOG artifact and the actual EEG signal. In [32], the kPCA is evaluated with two different kernel functions as polynomial function and RBF (radial basis function). However, the kPCA with RBF, 3rd order polynomial function is considered for comparison purpose.

4. Experimental results

The performance of the proposed VME-GMETV model is analyzed with following parameters.

(i) The RRMSE [21,26,27] is defined as

$$RRMSE = \frac{RMS(y(k) - \tilde{z}(k))}{RMS(y(k))} \quad (10)$$

where, $y(k)$ and $\tilde{z}(k)$ are clean EEG and denoised EEG respectively.

(ii) The CC [27] is given as

$$CC = \frac{\text{covariance}(y(k), \tilde{z}(k))}{\sigma_{y(k)}\sigma_{\tilde{z}(k)}} \quad (11)$$

where σ is standard deviation

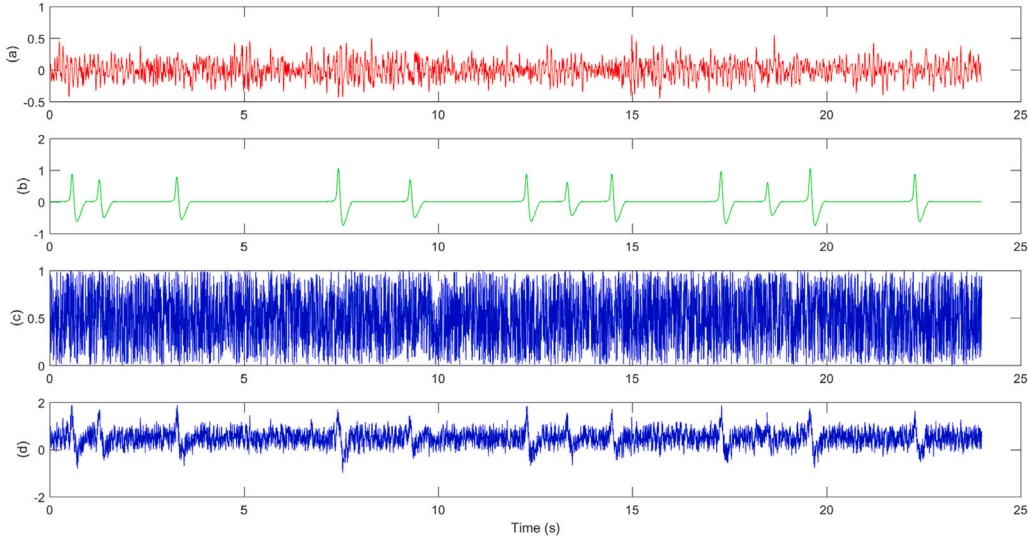


Fig. 2. (a) Clean EEG, (b) EOG artifact, (c) High frequency random signal and (d) Contaminated EEG with -2 dB.

(iii) The MAE is represented as [21,26]

$$MAE = \frac{\sum_{k=i}^j |p_{z(k)} - p_{z(k)}|}{i - j} \quad (12)$$

where, $p_{z(k)}$ and $p_{z(k)}$ are power spectrum of the denoised EEG and power spectrum of contaminated EEG respectively. The $i - j$ is the range of frequencies.

4.1. Synthetic EEG

The synthetic data is constructed as per the following mathematical Eq. (13).

$$z(k) = y(k) + a(k) + p(k) \quad (13)$$

The EOG artifact, $a(k)$ is collected from [37], in this the EOG artifact having various magnitudes and placed at different time spans as shown in Fig. 2.(b). The clean EEG, $y(k)$ is collected from [38], which contains 54 number of records with a sampling frequency 200 Hz as displayed in Fig. 2.(a) one of them. A high frequency random signal, $p(k)$ is considered as displayed in Fig. 2.(c) to maintain closeness with real time data. By adding all these three signals, the contaminated synthetic data, $z(k)$ is generated as displayed in Fig. 2.(d).

To make effective representation of proposed VME-GMETV denoising performance, the 24 s long synthetic contaminated EEG data is splits into eight segments of each 3-s duration. These segments are applied to VME algorithm, which extracts the desired mode as shown in Figs. 3 to 10(b). The desired mode is given to GMETV filter after estimation of local maxima, which extracts the actual artifact from the desired mode as shown in Figs. 3 to 10(c). Then, this actual artifact is subtracted from the desired mode yields residue of wanted EEG portion as displayed in Figs. 3 to 10(d). The final denoised EEG is obtained by adding this residue to the result of the subtraction of the contaminated segment and desired mode, which is depicted in Figs. 3 to 10(e). From the Figs. 3 to 10, it is observed that the contaminated EEG data having single instant, two instants and three instants of EOG artifacts, the proposed VME-GMETV method successfully removed multi-instants of EOG artifacts. To evolute the performance of existing and proposed methods, 19 number of records with 24 s duration having different signal to noise ratios (SNR) values of three sets as poor SNR (< -8 dB), mild SNR ($-8 \text{ dB} < \text{SNR} < 0$ dB) and good SNR (> 0 dB) are considered. The performance measures, $RRMSE$ and CC values are calculated for both proposed and existing methods and tabulated as shown in Tables 1–6.

Table 1

Proposed VME-GMETV model performance metrics for various SNRs.

S.no	Poor SNR		Mild SNR		Good SNR	
	CC	RRMSE	CC	RRMSE	CC	RRMSE
1	0.9324	0.2006	0.9594	0.1611	0.9670	0.1511
2	0.9258	0.2271	0.9627	0.1699	0.9713	0.1502
3	0.9295	0.2297	0.9618	0.1792	0.9700	0.1626
4	0.9161	0.2618	0.9541	0.2218	0.9643	0.2019
5	0.9354	0.2347	0.9630	0.1892	0.9708	0.1725
6	0.9327	0.2155	0.9639	0.1777	0.9715	0.1595
7	0.9496	0.1506	0.9664	0.1412	0.9712	0.1399
8	0.9476	0.1641	0.9642	0.1581	0.9696	0.1536
9	0.9504	0.1441	0.9684	0.0625	0.9734	0.1023
10	0.9418	0.1499	0.9611	0.1422	0.9678	0.1617
11	0.9258	0.2052	0.9595	0.1641	0.9681	0.1605
12	0.9213	0.2359	0.9602	0.1712	0.9690	0.1583
13	0.9248	0.2341	0.9654	0.1673	0.9735	0.1494
14	0.9260	0.2168	0.9641	0.1726	0.9726	0.1574
15	0.9443	0.1441	0.9663	0.1543	0.9745	0.1407
16	0.9408	0.1590	0.9642	0.1029	0.9704	0.0968
17	0.9289	0.2536	0.9564	0.2182	0.9650	0.2020
18	0.9321	0.2342	0.9588	0.1951	0.9665	0.1804
19	0.9499	0.1434	0.9591	0.1612	0.9648	0.1584
Average	0.9345	0.2002	0.9621	0.1637	0.9695	0.1557

From the Tables 1–6, it is noticed that the proposed method is outperformed than the existing methods and obtained the averaged CC is 0.9695 and averaged $RRMSE$ is 0.1557 at good SNR.

4.2. Real-time EEG

The real time EEG data is collected from the public domain BCI-IV study database [39–41]. It contains 64 number of records and each record having 23 number of channels. For experimentation purpose six segments (each segment having 2000 samples) are considered as shown in Fig. 11.

The segment-1 is applied to both existing and proposed methods and the denoised signals are displayed in Fig. 12. From Fig. 12, it is observed that the proposed VME-GMETV method effectively removed the artifact portion than the existing methods which is highlighted by the circle. To analyze the performance of existing and proposed methods numerically on real time EEG data, the performance measure, MAE is considered. In order to calculate the MAE , the denoised signals of proposed and existing methods PSD(power spectral density) plots are drawn and displayed in Fig. 13. From the PSD plots the MAE

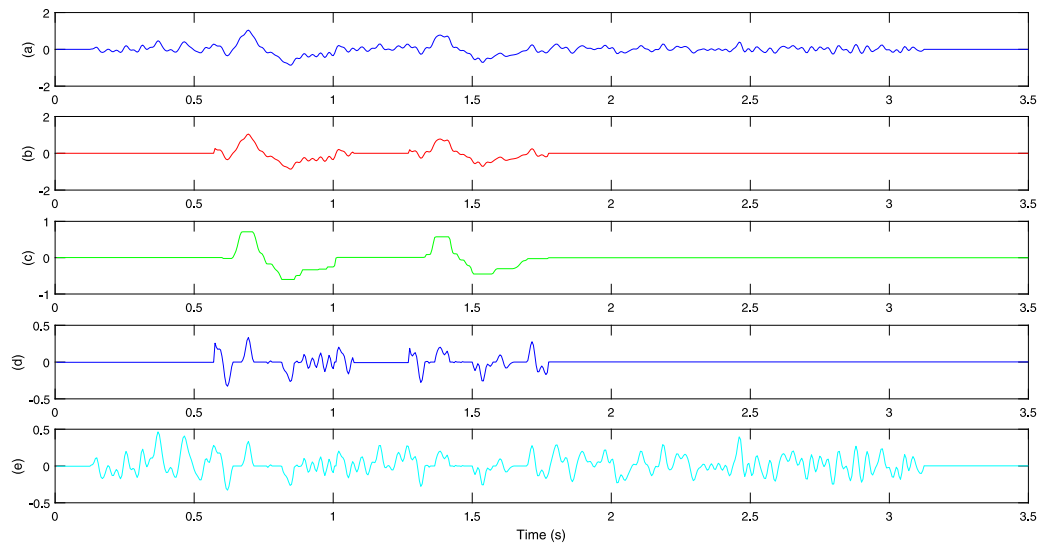


Fig. 3. (a) Contaminated EEG segment-1, (b) Desired mode, $x_d(k)$, (c) GMETV filtered output, $x_{filter}(k)$, (d) Residue of the wanted EEG, $\tilde{x}(k)$ and (e) Denoised EEG.

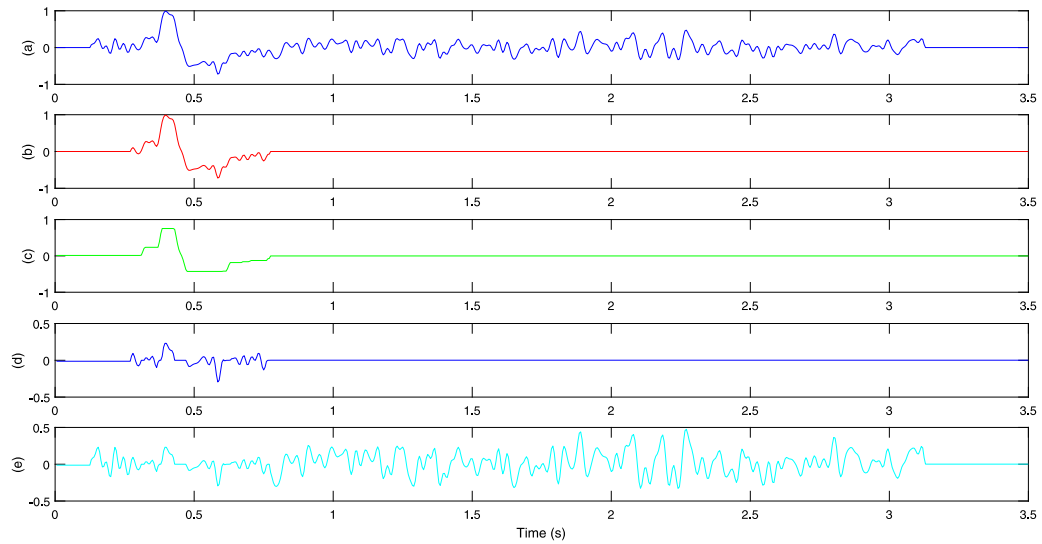


Fig. 4. (a) Contaminated EEG segment-2, (b) Desired mode, $x_d(k)$, (c) GMETV filtered output, $x_{filter}(k)$, (d) Residue of the wanted EEG, $\tilde{x}(k)$ and (e) Denoised EEG.

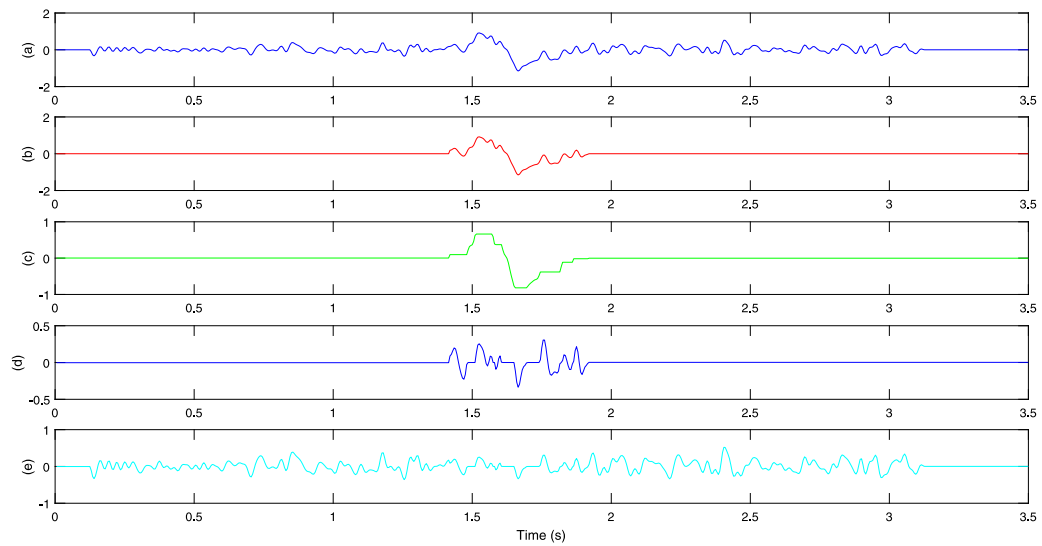


Fig. 5. (a) Contaminated EEG segment-3, (b) Desired mode, $x_d(k)$, (c) GMETV filtered output, $x_{filter}(k)$, (d) Residue of the wanted EEG, $\tilde{x}(k)$ and (e) Denoised EEG.

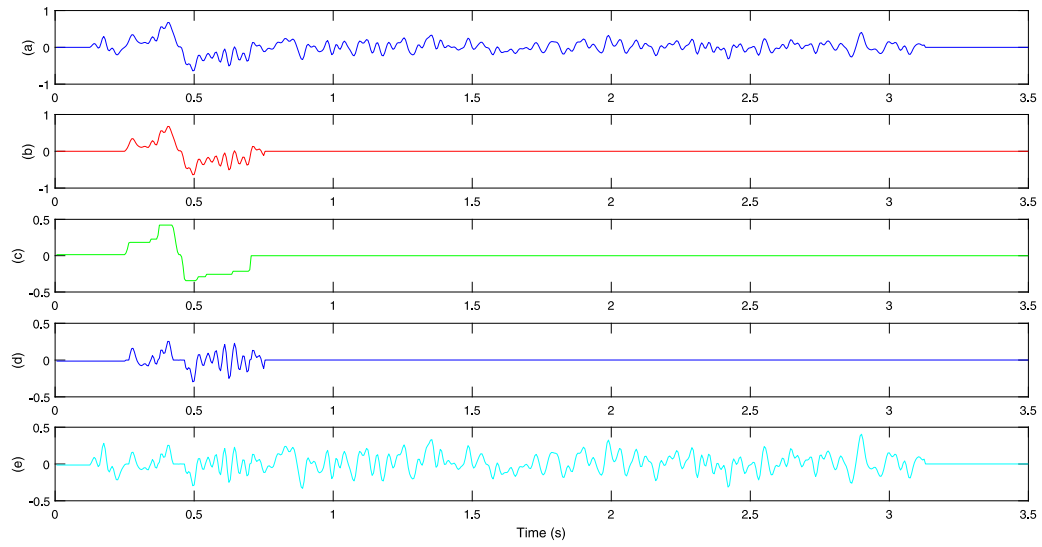


Fig. 6. (a) Contaminated EEG segment-4, (b) Desired mode, $x_d(k)$, (c) GMETV filtered output, $x_{filter}(k)$, (d) Residue of the wanted EEG, $\tilde{x}(k)$ and (e) Denoised EEG.

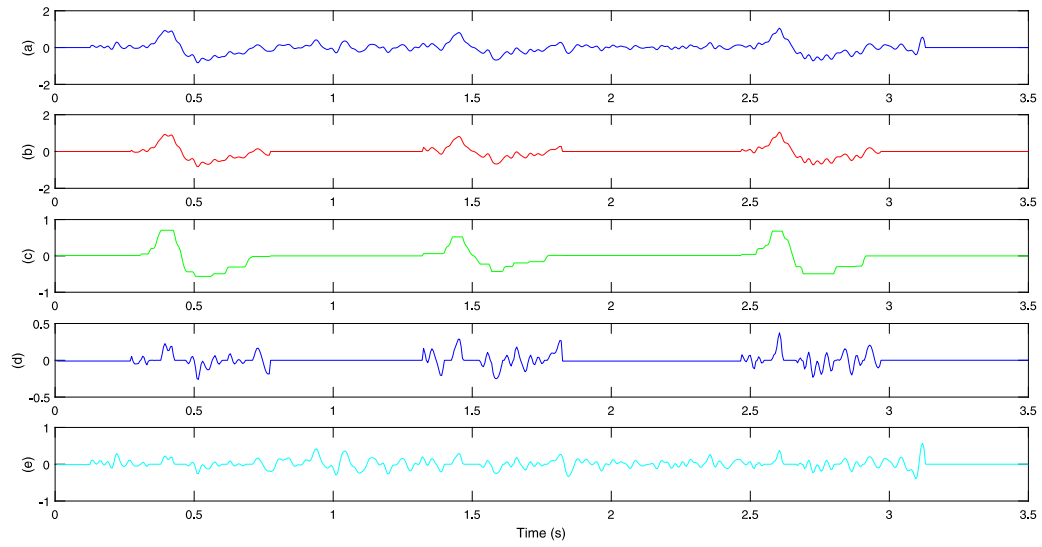


Fig. 7. (a) Contaminated EEG segment-5, (b) Desired mode, $x_d(k)$, (c) GMETV filtered output, $x_{filter}(k)$, (d) Residue of the wanted EEG, $\tilde{x}(k)$ and (e) Denoised EEG.

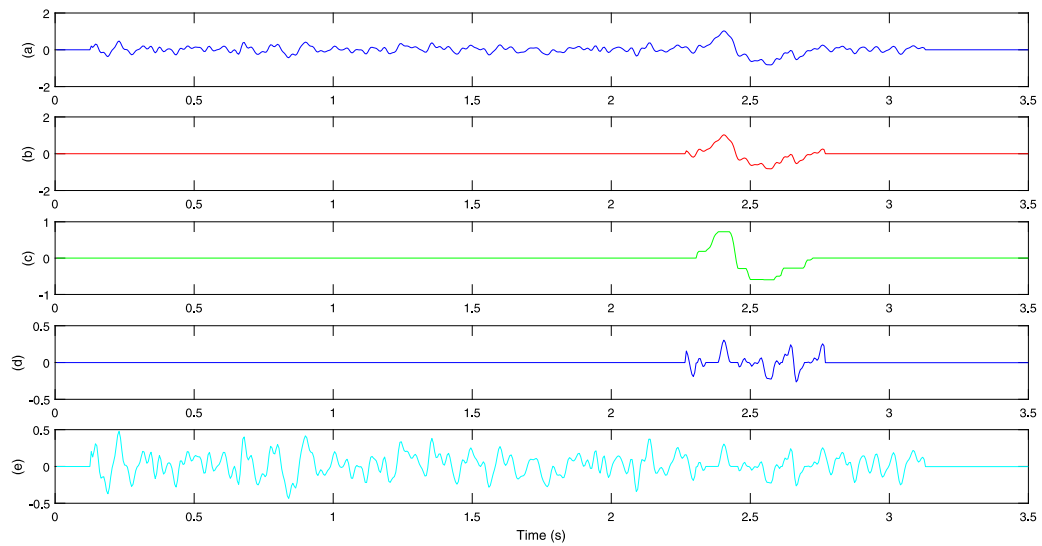


Fig. 8. (a) Contaminated EEG segment-6, (b) Desired mode, $x_d(k)$, (c) GMETV filtered output, $x_{filter}(k)$, (d) Residue of the wanted EEG, $\tilde{x}(k)$ and (e) Denoised EEG.

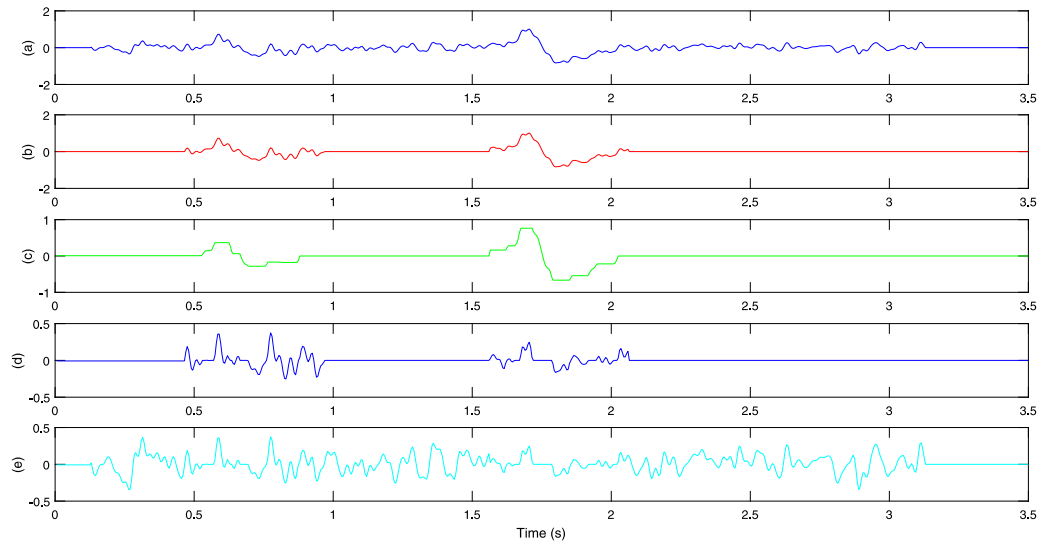


Fig. 9. (a) Contaminated EEG segment-7, (b) Desired mode, $x_d(k)$, (c) GMETV filtered output, $x_{filter}(k)$, (d) Residue of the wanted EEG, $\tilde{x}(k)$ and (e) Denoised EEG.

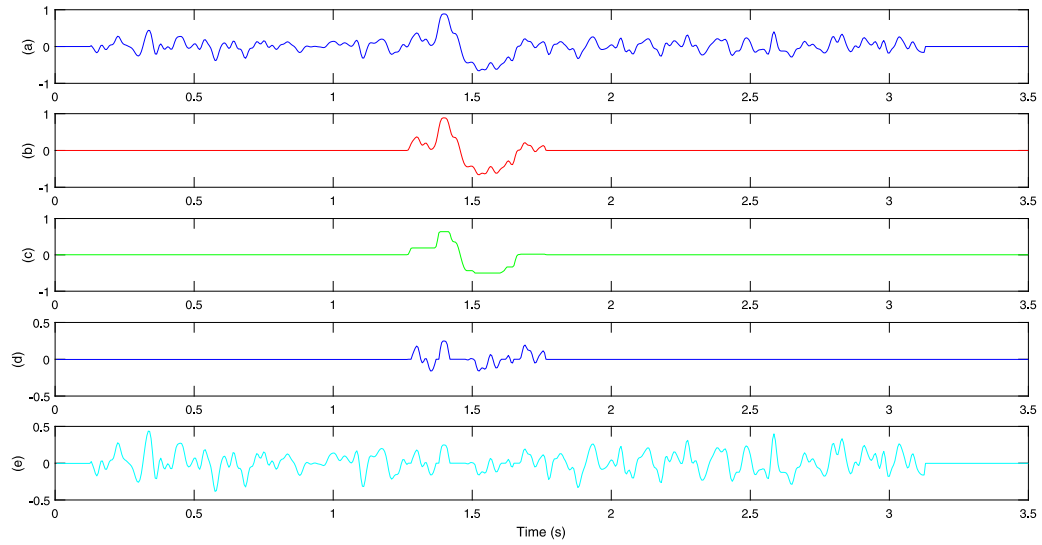


Fig. 10. (a) Contaminated EEG segment-8, (b) Desired mode, $x_d(k)$, (c) GMETV filtered output, $x_{filter}(k)$, (d) Residue of the wanted EEG, $\tilde{x}(k)$ and (e) Denoised EEG.

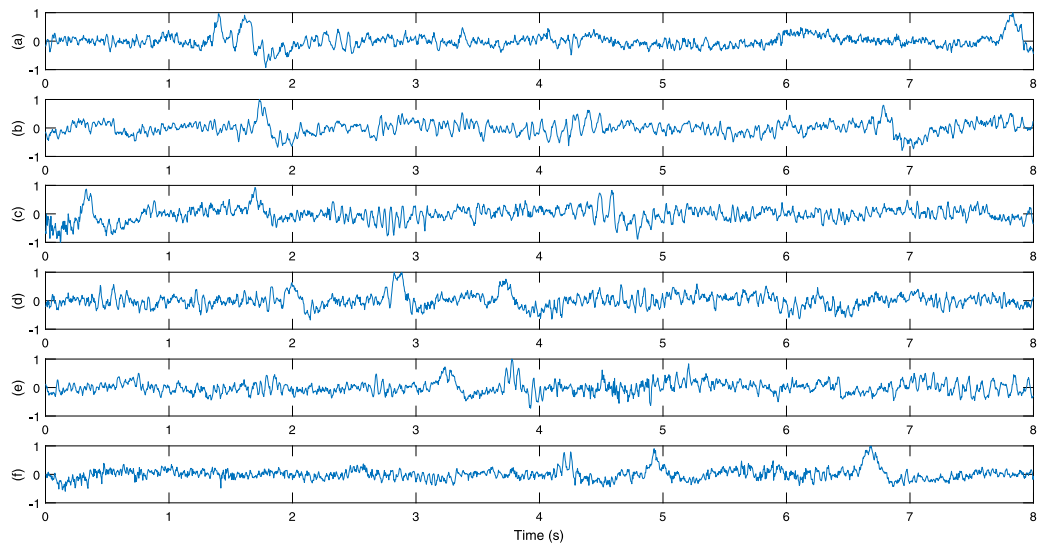


Fig. 11. (a) Segment-1, (b) Segment-2, (c) Segment-3, (d) Segment-4, (e) Segment-5 and (f) Segment-6.

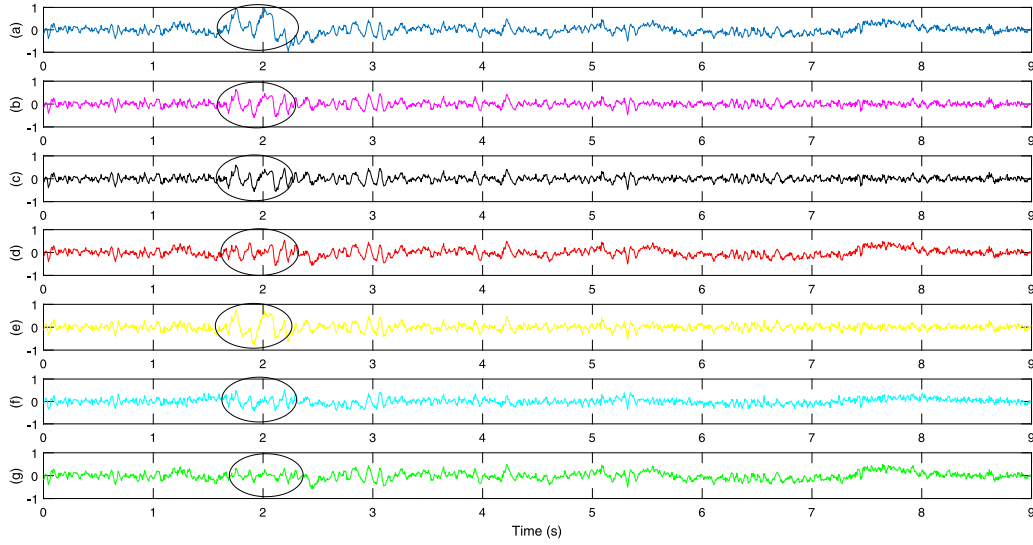


Fig. 12. (a) Real EEG segment-1, (b) Denoised EEG by Ov-ASSA-ANC, (c) Denoised EEG by SSA-ICA-SWT, (d) Denoised EEG by VME-DWT (e) Denoised EEG by EMD-Cross-correlation, (f) Denoised EEG by EEMD-kPCA and (g) Denoised EEG by proposed VME-GMETV.

Table 2

Existing VME-DWT [27] model performance metrics for various SNRs.

S.no	Poor SNR		Mild SNR		Good SNR	
	CC	RRMSE	CC	RRMSE	CC	RRMSE
1	0.9066	0.2704	0.9481	0.2155	0.9596	0.1970
2	0.8580	0.2908	0.9136	0.2317	0.9510	0.2111
3	0.9031	0.2694	0.9485	0.1903	0.9610	0.1698
4	0.8450	0.3089	0.9391	0.2369	0.9538	0.2137
5	0.9117	0.2148	0.9278	0.1886	0.9455	0.1783
6	0.8715	0.2557	0.9079	0.1907	0.9338	0.1740
7	0.9007	0.1777	0.9238	0.1667	0.9401	0.1611
8	0.9378	0.1966	0.9585	0.1769	0.9663	0.1693
9	0.8594	0.1938	0.8876	0.1718	0.9341	0.1639
10	0.9048	0.2174	0.9452	0.2011	0.9549	0.1908
11	0.8457	0.2719	0.9225	0.2205	0.9419	0.2024
12	0.8902	0.2717	0.9224	0.1631	0.9427	0.1438
13	0.8480	0.2570	0.9095	0.1945	0.9347	0.1770
14	0.8857	0.2806	0.9429	0.1890	0.9578	0.1685
15	0.8780	0.1855	0.9224	0.1513	0.9639	0.1443
16	0.9287	0.2303	0.9609	0.1989	0.9695	0.1857
17	0.9065	0.2654	0.9432	0.2290	0.9555	0.2131
18	0.9075	0.2459	0.9271	0.2251	0.9447	0.2119
19	0.9145	0.1821	0.9595	0.1923	0.9665	0.1879
Average	0.8897	0.2414	0.9321	0.1965	0.9514	0.1823

Table 3

Existing SSA-ICA-SWT [21] model performance metrics for various SNRs.

S.no	Poor SNR		Mild SNR		Good SNR	
	CC	RRMSE	CC	RRMSE	CC	RRMSE
1	0.7588	0.3063	0.8557	0.2556	0.8889	0.2369
2	0.7414	0.3278	0.8553	0.2646	0.8904	0.2439
3	0.7634	0.3015	0.8648	0.2264	0.8974	0.2048
4	0.7339	0.3350	0.8506	0.2601	0.8873	0.2365
5	0.7834	0.2476	0.8674	0.2119	0.8973	0.2004
6	0.7732	0.2749	0.8676	0.2196	0.8989	0.2046
7	0.9025	0.2335	0.9332	0.2359	0.9045	0.1879
8	0.9045	0.2469	0.9329	0.2461	0.9438	0.2422
9	0.8089	0.2020	0.8820	0.1886	0.9070	0.1827
10	0.7912	0.2139	0.8711	0.2028	0.9011	0.1966
11	0.7268	0.2973	0.8527	0.2417	0.8901	0.2228
12	0.7281	0.3056	0.8594	0.2183	0.8961	0.1986
13	0.7432	0.2735	0.8609	0.1984	0.8955	0.1823
14	0.7150	0.2937	0.8507	0.2083	0.8888	0.1898
15	0.7864	0.1793	0.8743	0.1608	0.9033	0.1603
16	0.8037	0.2172	0.8792	0.1906	0.9051	0.1823
17	0.7765	0.2985	0.8624	0.2479	0.8932	0.2301
18	0.7701	0.2768	0.8627	0.2464	0.8940	0.2315
19	0.8313	0.2549	0.9084	0.4145	0.9063	0.2158
Average	0.7812	0.2677	0.8732	0.2336	0.8994	0.2079

is evaluated for all the six segments in various bands like δ band (0.5–4 Hz), θ band (4–8 Hz), α band (8–13 Hz), β band (13–30 Hz) and γ band (30–100) and displayed in Table 7.

From the existing Ov-ASSA-ANC [26], the contaminated signal is divided into segments with a small portion of overlapping (5% of overlapping gives better performance as per the Ref. [11]). The individual segment is given to adaptive SSA. Here, the ASSA window length is considered 50 for effective performance. The first reconstructed component is considered as EOG artifact and fed to ANC as the reference input, which extracts the denoised signal. The denoised signal is displayed in Fig. 12(b). The PSD plot of the denoised signal is displayed in Fig. 13(b), and the MAE values of various bands are calculated and displayed in Table 7. From the Table 7, it is noticed that, the existing Ov-ASSA-ANC algorithm is removing EOG artifact along with wanted portions of EEG especially in α band, whose value is 0.0170. The disadvantage of this existing algorithm [26] is, it needs adjusting of overlapping portion for effective performance.

From the existing SSA-ICA-SWT [21], the single channel EEG is decomposed into several bands using SSA repeatedly. However, in this

paper five decomposition levels and window length 50 are considered as per the Ref. [21]. Then ICA is applied on these band of the signals, which will separate the hidden sources as in the form of independent components (IC). The soft thresholding is applied on the artifact IC through the SWT technique. Here, the SWT technique with db4 calculates the detailed and approximation coefficients of artifact IC and these coefficients are compared with soft thresholding. Those coefficients above soft thresholding are set to zero and remaining coefficients are considered as wanted component. The denoised signal is displayed in Fig. 12(c). The PSD plot of the denoised signal is displayed in Fig. 13(c), and the MAE values of various bands are calculated and displayed in Table 7. From the Table 7, it is observed that, the existing SSA-ICA-SWT algorithm is removing EOG artifact along with wanted portions of EEG especially in α band, whose value is 0.0108.

In the existing VME-DWT [27], the EOG artifact segment is identified by VME technique. Here, the VME technique parameters, compactness co-efficient, c is 3000 and center frequency, ω is 3 Hz are considered for the estimation of desired mode. Then, the identified EOG

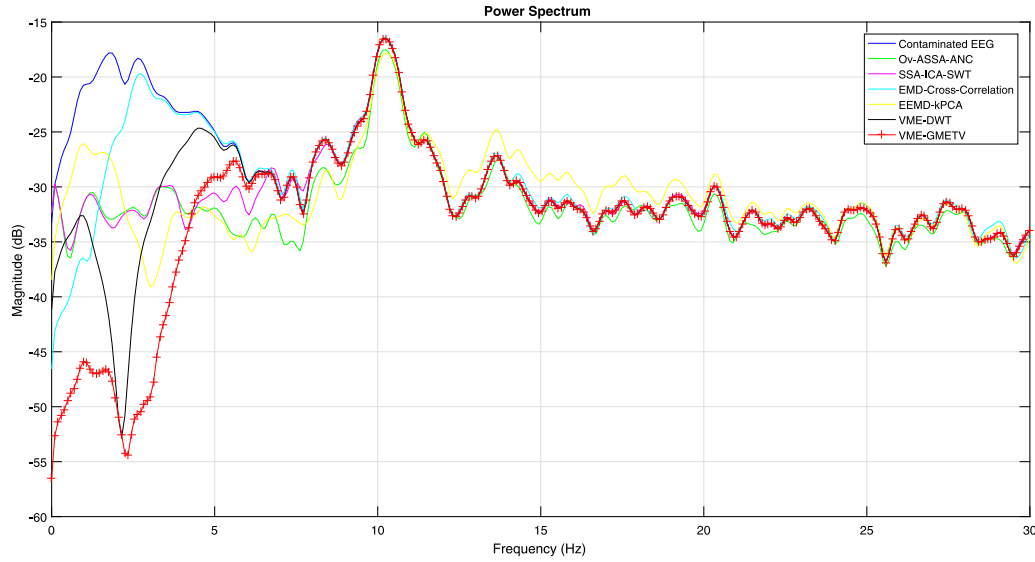


Fig. 13. Power spectral density plots of (a) Real time EEG (blue), (b) Denoised EEG by Ov-ASSA-ANC (green), (c) Denoised EEG by SSA-ICA-SWT (magenta), (d) Denoised EEG by EMD-Cross-correlation (cyan), (e) Denoised EEG by EEMD-kPCA (yellow), (f) Denoised EEG by VME-DWT (black) and (g) Denoised EEG by proposed VME-GMETV (red).

Table 4
Existing Ov-ASSA-ANC [26] model performance metrics for various SNRs.

S.no	Poor SNR		Mild SNR		Good SNR	
	CC	RRMSE	CC	RRMSE	CC	RRMSE
1	0.7141	0.3455	0.8279	0.2976	0.8683	0.2807
2	0.6967	0.3465	0.8243	0.2788	0.8669	0.2586
3	0.7175	0.3521	0.8334	0.2876	0.8728	0.2668
4	0.6851	0.3704	0.8170	0.3123	0.8615	0.2909
5	0.7381	0.3495	0.8349	0.2941	0.8718	0.2738
6	0.7267	0.3510	0.8370	0.2809	0.8743	0.2608
7	0.7735	0.3307	0.8500	0.2901	0.8816	0.2768
8	0.7863	0.3353	0.8604	0.2998	0.8893	0.2848
9	0.7656	0.3266	0.8503	0.2783	0.8823	0.2641
10	0.7418	0.3495	0.8384	0.3145	0.8742	0.2984
11	0.6736	0.3444	0.8171	0.2931	0.8629	0.2729
12	0.6792	0.3543	0.8280	0.2871	0.8721	0.2679
13	0.6893	0.3610	0.8232	0.2762	0.8667	0.2555
14	0.6600	0.3276	0.8151	0.2709	0.8618	0.2502
15	0.7302	0.3415	0.8382	0.2924	0.8761	0.2598
16	0.7540	0.3570	0.8469	0.2991	0.8804	0.2825
17	0.7334	0.3588	0.8296	0.3051	0.8671	0.2843
18	0.7251	0.3511	0.8321	0.9588	0.8704	0.2849
19	0.8005	0.3450	0.8585	0.3225	0.8857	0.3076
Average	0.7258	0.3473	0.8349	0.3284	0.8730	0.2748

Table 5
Existing EMD-Cross-Correlation [31] model performance metrics for various SNRs.

S.no	Poor SNR		Mild SNR		Good SNR	
	CC	RRMSE	CC	RRMSE	CC	RRMSE
1	0.6992	0.5003	0.7634	0.4971	0.8048	0.4672
2	0.7401	0.4477	0.7995	0.4312	0.8246	0.4140
3	0.6871	0.5151	0.7947	0.4456	0.8313	0.4221
4	0.7035	0.4824	0.7850	0.4646	0.8017	0.4288
5	0.7482	0.4579	0.7827	0.4265	0.8173	0.4171
6	0.7388	0.4601	0.7814	0.4497	0.8188	0.4135
7	0.7762	0.4274	0.8082	0.4152	0.8379	0.3822
8	0.7981	0.4531	0.8190	0.4005	0.8208	0.3219
9	0.7009	0.5457	0.7408	0.5144	0.7671	0.5058
10	0.7214	0.4525	0.7936	0.4490	0.8335	0.4266
11	0.7456	0.6621	0.7545	0.5052	0.8037	0.4637
12	0.6852	0.4727	0.7985	0.4335	0.8280	0.4190
13	0.7759	0.6203	0.7559	0.4991	0.7927	0.4781
14	0.7807	0.5340	0.7679	0.4701	0.8005	0.4570
15	0.7488	0.4645	0.7781	0.4191	0.8032	0.3918
16	0.7623	0.4194	0.8108	0.3995	0.8442	0.3785
17	0.7312	0.5087	0.7612	0.4664	0.7968	0.4886
18	0.7463	0.4807	0.7832	0.4734	0.8053	0.4184
19	0.7948	0.4503	0.8039	0.4498	0.8264	0.3848
Average	0.7418	0.4924	0.7833	0.4532	0.8136	0.4252

segment is applied to DWT technique with db4, where the skewness-based thresholding is performed. The denoised signal and its PSD plots are displayed in Figs. 12(d), 13(f) respectively. The MAE values of various bands are calculated and displayed in Table 7 and it is observed that the α band MAE is 0.0027. The existing methods in [21,27], both are used SWT/DWT with thresholding. The performance of these [21, 27] methods are depended on the selection of mother wavelet as well as decomposition levels. However, the thresholding introduces loss of wanted component due to non-stationary behavior of EEG signals, which leads the performance degradation [25,28,29].

From the existing EMD-Cross-correlation [31], the single channel contaminated EEG signal is applied to EMD technique, it was decomposed into seven number of intrinsic mode functions (IMFs). The EOG artifact contaminated IMFs are identified based on the statistical parameters, Variance and Skewness. As per the authors in [31], the IMFs having high values of variance and skewness are considered as EOG artifact contaminated IMF. To conform this another statistical parameter cross-correlation coefficient is calculated between the EOG

artifact contaminated IMF and the contaminated EEG signal. If the calculated cross-correlation is greater than the threshold cross-correlation that IMFs are removed and the remaining IMFs are combined, it results the denoised EEG, which is displayed in Fig. 12(e). The PSD plot of this denoised signal is displayed in Fig. 13(d). The threshold cross-correlation value is adjusted such that which gives the low RRMSE value. In this paper the threshold cross-correlation value is considered 0.6, where, the authors in [31] is 0.5. The MAE values of various bands of denoised signal are calculated and displayed in Table 7. From the Table 7, it is observed that the removal of EOG artifact is poorly done compared to other existing and proposed methods, especially the α band MAE is 0.0216.

From the existing EEMD-kPCA [32], the single channel interfered EEG signal is decomposed into seven number of IMFs using EEMD technique. These IMFs are applied to kPCA, which separates the EOG artifact and the actual EEG signal. In [32], the kPCA is evaluated with two different kernel functions as polynomial function and RBF (radial basis function). However, the kPCA with RBF, third-order polynomial

Table 6
Existing EEMD-kPCA [32] model performance metrics for various SNRs.

S.no	Poor SNR		Mild SNR		Good SNR	
	CC	RRMSE	CC	RRMSE	CC	RRMSE
1	0.8011	0.5367	0.8324	0.5162	0.8555	0.4716
2	0.7539	0.5396	0.8416	0.4705	0.8620	0.4494
3	0.7891	0.5482	0.8590	0.4634	0.8821	0.4179
4	0.7602	0.5209	0.8176	0.5007	0.8460	0.4759
5	0.7789	0.5069	0.8567	0.4550	0.8626	0.4502
6	0.8223	0.5105	0.8465	0.4765	0.8373	0.4776
7	0.8143	0.4517	0.8604	0.4354	0.8896	0.4353
8	0.8554	0.4650	0.8747	0.4462	0.8988	0.3740
9	0.8398	0.5071	0.8471	0.4976	0.8643	0.4571
10	0.8529	0.4567	0.8663	0.4470	0.8698	0.3880
11	0.6875	0.4806	0.8208	0.4543	0.8434	0.4392
12	0.7599	0.4863	0.8647	0.4449	0.8798	0.4168
13	0.7741	0.5281	0.8248	0.5094	0.8655	0.4705
14	0.6977	0.5029	0.8446	0.4765	0.8594	0.4569
15	0.7611	0.4712	0.8400	0.4752	0.8572	0.4699
16	0.7983	0.4579	0.8597	0.4459	0.8862	0.4161
17	0.7800	0.5452	0.8321	0.5135	0.8502	0.4867
18	0.8161	0.5204	0.8446	0.4971	0.8531	0.4469
19	0.8244	0.4445	0.8823	0.4156	0.9011	0.3899
Average	0.7877	0.4990	0.8482	0.4706	0.8665	0.4416

Table 7
MAE of PSD plots in various frequency bands.

Method	Epoch	δ band	θ band	α band	β band	γ band
Ov-ASSA-ANC [26]	1	0.1381	0.0305	0.0282	0.0069	0.0067
	2	0.1292	0.0335	0.0120	0.0064	0.0068
	3	0.0853	0.0409	0.0189	0.0052	0.0065
	4	0.1298	0.0358	0.0236	0.0056	0.0304
	5	0.1289	0.0386	0.0149	0.0064	0.0331
	6	0.1041	0.0340	0.0041	0.0068	0.0074
Average		0.1192	0.0355	0.0170	0.0062	0.0151
SSA-ICA-SWT [21]	1	0.1379	0.0260	0.0103	0.0012	4.2211e-04
	2	0.1338	0.0238	0.0104	0.0036	4.3751e-04
	3	0.1346	0.0336	0.0096	0.0013	3.9239e-04
	4	0.1370	0.0347	0.0122	0.0045	0.0076
	5	0.1432	0.0349	0.0136	0.0027	0.0073
	6	0.1232	0.0309	0.0084	0.0023	3.8270e-04
Average		0.1349	0.0307	0.0108	0.0026	0.0028
EMD-Cross correlation [31]	1	0.1001	0.0176	0.0117	0.0036	0.0143
	2	0.0956	0.0269	0.0177	0.0049	0.0138
	3	0.0993	0.0119	0.0201	0.0049	0.0133
	4	0.1045	0.0214	0.0402	0.0039	0.0129
	5	0.0992	0.0162	0.0285	0.0050	0.0144
	6	0.0920	0.0205	0.0116	0.0044	0.0128
Average		0.0984	0.0191	0.0216	0.0045	0.0136
EEMD-kPCA [32]	1	0.1389	0.0368	0.0047	0.0111	0.0146
	2	0.1348	0.0363	0.0142	0.0144	0.0136
	3	0.1342	0.0406	0.0057	0.0063	0.0134
	4	0.1379	0.0249	0.0200	0.0112	0.0132
	5	0.1407	0.0229	0.0012	0.0177	0.0156
	6	0.1328	0.0258	0.0077	0.0092	0.0131
Average		0.1366	0.0312	0.0089	0.0117	0.0139
VME-DWT [27]	1	0.1497	0.0237	0.0027	9.6900e-04	2.3487e-04
	2	0.1430	0.0186	0.0021	0.0014	3.9393e-04
	3	0.1353	0.0267	0.0022	0.0010	3.5242e-04
	4	0.1464	0.0292	0.0033	0.0017	4.2762e-04
	5	0.1436	0.0174	0.0038	0.0014	5.8303e-04
	6	0.1330	0.0199	0.0020	0.0011	3.3686e-04
Average		0.1418	0.0226	0.0027	0.0013	3.8812e-04
Proposed VME-GMETV	1	0.1501	0.0203	0.0018	1.2057e-04	1.9352e-04
	2	0.1451	0.0019	3.3596e-04	1.5389e-04	1.3544e-04
	3	0.1440	0.0183	5.9064e-05	1.8940e-04	4.4731e-05
	4	0.1473	0.0172	3.1639e-04	3.9793e-05	1.8070e-04
	5	0.1504	0.0105	4.2910e-04	1.1789e-04	2.3802e-04
	6	0.1405	0.0073	1.0230e-04	1.5103e-04	1.0728e-04
Average		0.1462	0.0126	5.0714e-04	1.2876e-04	1.4995e-04

function is considered for comparison purpose. The denoised and its PSD plots are displayed in Figs. 12(f) and 13(e). The MAE values of various bands of denoised signal are calculated and displayed in Table 7. From the Table 7, it is observed that the removal of EOG artifact is good compared to other existing Ov-ASSA-ANC [26] and SSA-ICA-SWT [21], but it is failed to retrieve other band of signals, especially the α band MAE is 0.0089 than the proposed method.

In the proposed VME-GMETV technique, the VME experimental parameters, compactness co-efficient, c is 3500 and center frequency, ω is 2.8 Hz are considered for the good estimation of desired mode. In the GMETV filter the regularization parameter is considered as 1.15 for better estimation of EOG artifact. By using these parameters, the denoised signal is obtained and its PSD plots are displayed in Figs. 12(g), 13(g) respectively. The MAE values of various bands of denoised signal are calculated and displayed in Table 7.

From the Table 7, it is noticed that the proposed method effectively removed the EOG artifact in δ band and restoring the wanted portions in remaining bands, θ band, α band, β band and γ bands than the existing methods. Especially, it is very important for any denoised algorithm to restore the α band component, which is crucial for BCI applications. From the Table 7, it is observed the proposed method is accurately restored the α band component with least MAE, $5.0714e-04$ than the other existing methods, hence the proposed method suitable for BCI applications.

The novelty of the proposed work is as follows.

- (i). It requires less number decomposition levels, where the other existing methods need a greater number of decompositions, like EMD [31], EEMD-kPCA [32], SSA-ICA-SWT [21]. It means the complexity of the proposed algorithm is reduced.
- (ii). It is robust algorithm, which does not require any reference input, where the existing Ov-ASSA-ANC [26] required.
- (iii). Its performance is does not depends on number of decomposition levels as well as mother wavelets, where as the existing methods EMD [31], EEMD-kPCA [32], SSA-ICA-SWT [21], VME-DWT [27].

5. Conclusion

In this paper a novel framework is proposed as Variational Mode Extraction (VME) method is integrated with Generalized Moreau Envelope Total Variation (GMETV) technique to filter the EOG artifacts. Initially, the contaminated EEG is applied to the VME method which extracts the EOG artifact segment. Then, this EOG artifact segment is given to the GMETV technique and it separates the wanted component of the EEG residue from the EOG artifact segment. The EEG residue is added back to the non EOG artifact segment signal which gives the final denoised EEG. Matlab simulations are done on both simulated and real time EEG databases to evaluate the proposed VME-GMETV technique performance in terms of $RRMSE$, CC and MAE . From the experimental results, the proposed technique is outperformed the existing methods by averaged low $RRMSE$ (0.1557), MAE ($5.0714e-04$) and averaged high CC (0.9695). The proposed method is not only removed EOG artifact effectively but also restored the α band component accurately with least MAE than the other existing methods, hence the proposed method suitable for BCI applications.

CRedit authorship contribution statement

Sayed Khasim Noorbasha: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Investigation. **Kethepalli Mallikarjuna:** Supervision, Editing.

Declaration of competing interest

The authors declare there are no competing interests.

Data availability

Data will be made available on request.

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